Arfaa Mumtaz

780-934-8623 | arfaamumtaz@hotmail.com | https://arfaa.ca | https://linkedin.com/in/arfaamumtaz | Edmonton, AB

PROFESSIONAL SUMMARY

Full-stack developer specializing in React, Next.js, Node.js, and PostgreSQL. Built and deployed **production systems**, including a lead-generation platform generating **78** qualified buyer inquiries. Ranked **top 1%** nationally in Cyber Skyline NCL. Experience spans AI-integrated systems, embedded hardware/software integration, and security-focused applications. Seeking junior software or data roles.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C#, C++, Bash
Frontend: React, Next.js, Tailwind CSS, HTML/CSS
Backend & DB: Node.js, PostgreSQL, SQLite, Supabase, MongoDB
Cloud / DevOps: Git, GitHub Actions, Docker, Vercel, GCP, Cloudflare
AI / Data: Gemini API, RAG, Pandas, NumPy, Regression Modeling

PROJECTS

AB AutoFinder - Full-Stack Lead-Generation Platform *Feb 2026 – Mar 2026*
https://abautofinder.ca · Next.js · PostgreSQL · Supabase Auth · Vercel · Cloudflare · Resend

- Shipped a production lead capture system (Next.js, PostgreSQL), generating **1,630+ users** and **78 qualified buyer inquiries**
- Built **SSR-based** intake pipeline with authentication and **automated email workflows**, converting structured buyer input into actionable sales leads
- Deployed secure backend behind Cloudflare, handling **29,650+ requests** (**~1.37 GB traffic**) with hardened session and query protection.

Agrilo - Agricultural IoT Framework (Funded Capstone) *Sept 2024 – Dec 2024*
MacEwan University · .NET MAUI · C# · Python · Gemini API · RAG · SQLite · AES-256

- Built and deployed a **cross-platform** agronomic system processing multi-factor inputs (geolocation, nutrient levels, crop type) to generate **data-driven recommendations**
- Modeled soil nutrient data (PPM) from RGB sensor inputs using **regression techniques**, supporting lower-cost alternatives to lab-based workflows valued at up to **~$15,000** per season
- Designed normalized relational database schema (**9+ tables**) enabling scalable storage/querying of structured agricultural test data
- Integrated **LLM-based** recommendation engine (**Gemini**) with **RAG**, reducing hallucination risk through retrieval from structured domain data

UAV-Based Agronomic Sampling System (USRI-Funded Research) *Jan 2025 – Jun 2025*
MacEwan University · Python · C++ · MAVLink · Pixhawk 6C · Raspberry Pi 4B · ArduPilot

- Led end-to-end development of an autonomous UAV sampling system (2-person team), integrating **embedded systems**, flight control, and hardware design into a **field-tested proof-of-concept** with ~30 min flight capacity
- Integrated **Raspberry Pi 4B** with Pixhawk 6C via **MAVLink** (UART/TELEM2), enabling autonomous GPS waypoint navigation, real-time telemetry processing, and sensor-driven sampling validated through repeated flight tests
- Validated waypoint navigation, telemetry flow, and flight stability through repeated field testing, resolving GPS interference and calibration issues

PROFESSIONAL EXPERIENCE

Research Assistant - UAV Agronomic Sampling System *Mar 2025 – Jun 2025*
MacEwan University, Edmonton, AB (Contract)

- **Led** the UAV project under MacEwan contract, partnering with **CS and Chemistry faculty** on hardware/software integration.

Security Systems Analyst Intern *Sept 2024 – Jan 2025*
Alternative Legal Services Firm Inc., Edmonton, AB (Contract / Internship)

- Executed security audit across 12+ endpoints (network, hardware, M365); delivered formal incident response plan **adopted by the firm**.
- Deployed BitLocker, enforced 2FA, configured Windows Defender, and migrated email domains to close authentication vulnerabilities.

EDUCATION

B.Sc. Computer Science - Systems & Information Security · Minor: Accounting *Sept 2021 – Jun 2025*
MacEwan University, Edmonton, AB · GPA: 3.40 / 4.00 (final 30 credits)

- Cyber Skyline NCL, Nov 2024: Individual **top 2%** (199th / 8,484 competitors); Team **top 1%** (72nd / 4,893 teams).